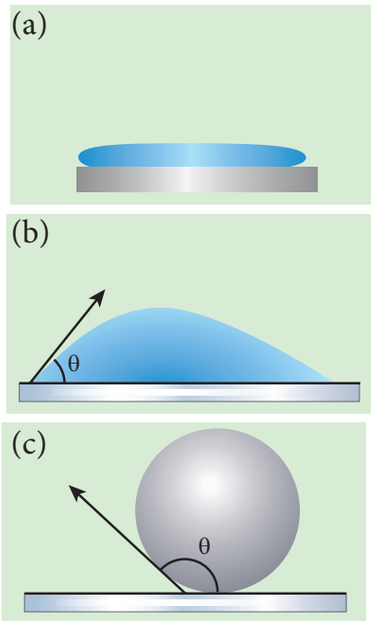


Figure 7.30 (a) water on silver surface (b) glass plate on water (c) glass on mercury

Practical applications of capillarity

- Due to capillary action, oil rises in the cotton within an earthen lamp. Likewise, sap rises from the roots of a plant to its leaves and branches.
- Absorption of ink by a blotting paper
- Capillary action is also essential for the tear fluid from the eye to drain constantly.
- Cotton dresses are preferred in summer because cotton dresses have fine pores which act as capillaries for sweat.

7.5.7 Surface Tension by capillary rise method

The pressure difference across a curved liquid-air interface is the basic factor behind the rising up of water in a narrow tube (influence of gravity is ignored). The capillary rise is more dominant in the case of very fine tubes. But this phenomenon is the outcome of the force of surface tension. In order to arrive a relation between the capillary rise (h) and surface tension (T), consider a capillary tube which is held vertically in a beaker containing water; the water rises in the capillary tube to a height h due to surface tension (Figure 7.31).

The surface tension force F_T, acts along the tangent at the point of contact downwards and its reaction force upwards. Surface tension T, is resolved into two components i) Horizontal component $T\sin\theta$ and ii) Vertical component $T\cos\theta$ acting upwards, all along the whole circumference of the meniscus.

$$\text{Total upward force} = (T\cos\theta)\,(2\pi r) = 2\pi r T\cos\theta$$

where θ is the angle of contact, r is the radius of the tube. Let ρ be the density of water and h be the height to which the liquid rises inside the tube. Then,

$$\begin{pmatrix}\text{the volume of}\\ \text{liquid column in}\\ \text{the tube,V}\end{pmatrix} = \begin{pmatrix}\text{volume of the}\\ \text{liquid column of radius r}\\ \text{height h}\end{pmatrix} + \begin{pmatrix}\text{volume of liquid of radius r}\\ \text{and height r - Volume of the}\\ \text{hemisphere of radius r}\end{pmatrix}$$

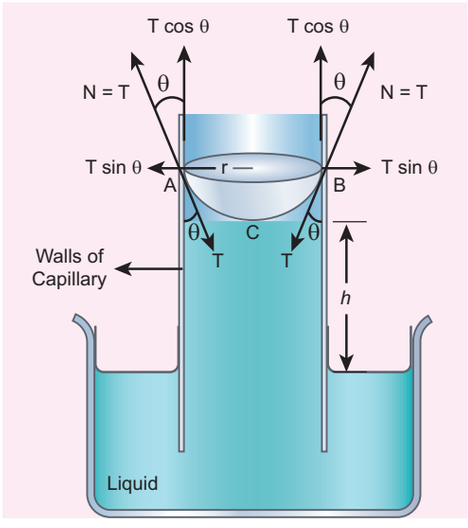


Figure 7.31 Capillary rise by surface tension

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